

Show Work! Point deductions taken if work not shown

1. Convert 1011_2 to decimal

$$2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$8 + 0 \times 4 + 1 \times 2 + 1 \times 1$$

$$8 + 0 + 2 + 1 = 11$$

2. Convert 111011_2 to decimal

$$2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$32 + 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$$

$$32 + 16 + 8 + 0 + 2 + 1 = 59$$

3. Convert 1011101_2 to decimal

$$2^6 + 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$64 + 0 \times 32 + 1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1$$

$$64 + 0 + 16 + 8 + 4 + 0 + 1 = 93$$

4. Convert 111000_2 to decimal

$$2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$32 + 1 \times 16 + 1 \times 8 + 0 \times 4 + 0 \times 2 + 0 \times 1$$

$$32 + 16 + 8 + 0 + 0 + 0 = 56$$

5. Convert 1100011_2 to decimal

$$2^6 + 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$64 + 1 \times 32 + 0 \times 16 + 0 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$$

$$64 + 32 + 0 + 0 + 0 + 2 + 1 = 99$$

6. Convert 11111111_2 to decimal

$$2^7 + 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$128 + 1 \times 64 + 1 \times 32 + 1 \times 16 + 1 \times 8 + 1 \times 4 + 1 \times 2 + 1 \times 1$$

$$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$$

7. Convert 10_2 to decimal

$$2^1 + 0 \times 2^0$$

$$2 + 0 \times 1$$

$$2 + 0 = 2$$

8. Convert 10_{16} to decimal

$$16^1 + 0 \times 16^0$$

$$16 + 0 \times 1$$

$$16 + 0 = 16$$

9. Convert AB_{16} to decimal

$$A = 10, B = 11$$

$$10 \times 16^1 + 11 \times 16^0$$

$$10 \times 16 + 11 + 1$$

$$160 + 11 = 171$$

10. Convert $F0_{16}$ to decimal

$$F = 15$$

$$15 \times 16^1 + 0 \times 16^0$$

$$15 \times 16 + 0 \times 1$$

$$240 + 0 = 240$$